

How much productivity do we need to reduce the workweek?

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Abstract

Brazil proposes to reduce the formal workweek from 44 to 36 hours (PEC 8/2025). This paper builds a structural partial-equilibrium model with a CES aggregator over formal and informal effective labor, calibrated to Brazilian microdata, and quantifies the short-run output cost and the one-off TFP gain that would offset it. The required gain is $A_{\text{req}} \approx 6.6\%$ under the preferred specification, with a conservative upper bound a couple of percentage points higher. The figure is high relative to the post-1990 Brazilian TFP series, in which productivity fell on average and rose only modestly in the best post-Real-Plan decade. The CES substitution elasticity above unity and a moderate informal efficiency parameter, identified from the observed informal-to-formal wage ratio, push marginal workers toward informality rather than absorbing the hours loss intensively. Aggregate informality rises rather than falls. Welfare peaks inside the model at a cap above 40 hours and turns negative further down. Small firms and agriculture bear disproportionate costs.

Keywords: working hours, informality, total factor productivity, CES aggregator, Brazil, welfare

JEL: E24, J22, J46, O17, O47

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1 Introduction

Brazil is debating an eight-hour cut to its standard workweek. PEC 8/2025 would lower the cap from 44 to 36 hours and bind roughly two thirds of formal employees at once. Two numbers anchor the policy discussion. The first is the output removed on impact. The second is the one-off productivity gain required to restore it.

The Brazilian setting complicates both answers. Informality is 37.8% under the narrow PNAD definition and 44.2% under the broad IBGE one, ranging from 41% in services to 72% in agriculture. Standard treatments of hours reform model a fully formal economy (Cacciatore et al., 2016) or treat informality as a static size distortion (Ulyssea, 2018). Neither has been applied to a large cap cut.

This paper builds a static partial-equilibrium model disciplined by Brazilian micro-data. A CES aggregator over formal and informal effective labor governs reallocation. A fatigue function $e(h)$ with peak at 40 hours maps the cap into per-hour productivity. Two firm-size groups generate differential incidence and GHH preferences support welfare. The substitution elasticity is pinned to the Brazilian formal–informal wage premium of Meghir et al. (2015), giving $\sigma_{\text{sub}} = 1.326$ with interval $[1.116, 1.469]$ at $(\omega, \eta_I) = (0.622, 0.40)$.

The central object is A_{req} , the one-off TFP gain that restores pre-reform output. Under the preferred flat-below specification $A_{\text{req}} = 6.63\%$. Under the conservative symmetric specification $A_{\text{req}} = 8.18\%$. The joint (σ, ω, η_I) envelope is $[5.62\%, 7.48\%]$ under the preferred specification and $[6.93\%, 9.23\%]$ under the conservative one. At Brazil’s best post-1990 decade (PWT `rtfpna`, +0.26% per year, 2003–2013) accumulating A_{req} takes 25–31 years. A stress test that layers a 1–2% post-reform TFP contraction raises the conservative A_{req} to 9.28–10.39%.

Aggregate informality rises by +1.6–1.9 pp. With $\sigma > 1$ and $\eta_I = 0.40$, formal and informal labor are imperfect substitutes, so the cap pushes marginal formal workers toward the informal sector instead of being absorbed by intensive adjustment alone. Welfare peaks near a 41-hour cap and turns negative below 38.

The paper makes three contributions. First, it embeds a formal–informal reallocation margin into a structural analysis of hours reform and shows the margin is modest under wage-premium-disciplined parameters. Second, it proposes A_{req} as a benchmark commensurable with the historical TFP record. Third, it provides a wage-premium-conditional interval for σ_{sub} that the reader can reassess.

Scope is limited. The framework is static and partial equilibrium. It omits capital deepening, general-equilibrium price adjustment, worker selection, and work intensification. Evidence from Portugal 1996 (Asai et al., 2024) and Brazil 1988 (Gonzaga et al., 2003) enters as a directional plausibility check rather than an identified lower bound for Brazil’s effective A_{req} .

Section 2 presents facts, model, and calibration. Section 3 validates the fit. Section 4

reports the counterfactual, the TFP benchmark, and the stress test. Section 5 draws policy implications.

2 Brazilian facts and framework

Four Brazilian facts motivate the model and discipline the calibration. First, formal hours are concentrated at the statutory maximum. DIEESE (2024) reports 8.5% of formal workers at 36 hours, 26.9% at 40 hours, and 64.6% at or near 44 hours, with an average of 42.2 hours per week. PNAD Contínua Q4 2024 habitual hours give a flatter distribution with an average of 41.2. The majority of formal workers operate at or above 40 hours, the margin a 36-hour cap would bind.

Second, informality is large and dispersed. The national rate is 37.8% (narrow) or 44.2% (broad). The sectoral range is 71.8% in agriculture, 44.4% in industry, and 41.2% in services (Table 1).

Third, the formal–informal wage premium is positive and moderate. Meghir et al. (2015) document a conditional wage premium $R \in [1.15, 1.55]$, the ratio of mean formal hourly wages to mean informal hourly wages among otherwise comparable workers. The calibration below uses this premium to pin down the CES substitution elasticity σ_{sub} .

Fourth, Brazilian TFP has been stagnant or declining throughout the post-1990 period. The Penn World Table 10.01 `rtfpna` series shows an annualized decline of -0.6% per year over 1990–2019 and growth of only $+0.26\%$ per year in the best post-Real Plan decade (2003–2013). The Brazilian TFP benchmark in Section 4 uses this series as the anchor for A_{req} .

Table 1: Sectoral and national labor-market moments, PNAD Contínua Q4 2024.

Sector	λ_s	Informality (broad)	Avg. hours (formal)	θ_{36}	θ_{40}	θ_{44}
Agriculture	0.076	71.8%	44.1	0.101	0.269	0.630
Industry	0.205	44.4%	42.5	0.048	0.417	0.535
Services	0.720	41.2%	40.8	0.171	0.410	0.419
National	1.000	44.2%	41.2	0.143	0.406	0.451

Notes: λ_s is the sector’s share of national employment. Informality uses the broad IBGE definition (VD4009 combined with V4017 CNPJ indicator). “Avg. hours” is the mean weekly hours of *formal* workers (V4039, habitual). θ_{36} , θ_{40} , θ_{44} are the shares of formal workers in each contracted-hours bin. *Source:* PNAD Contínua Q4 2024, IBGE microdata, survey-weighted, $N = 209,311$.

A representative firm in group $g \in \{\text{Small}, \text{Large}\}$ (small: at most 49 employees; large: 50 or more) produces output Y_g from capital K_g and effective labor L_g using a Cobb–Douglas technology,

$$Y_g = A \cdot K_g^\alpha \cdot L_g^{1-\alpha}, \quad (1)$$

where A is aggregate productivity and α is the capital share of income. Effective labor

L_g combines formal (F) and informal (I) workers through a CES aggregator,

$$L_g = [\omega L_{F,g}^\rho + (1 - \omega) L_{I,g}^\rho]^{1/\rho}, \quad \rho = (\sigma_{\text{sub}} - 1)/\sigma_{\text{sub}}, \quad (2)$$

where ω is the CES weight on formal labor and σ_{sub} is the elasticity of substitution between formal and informal labor. When $\sigma_{\text{sub}} < 1$ the two inputs are complements, and raising the cost of formal labor does not trigger a large reallocation toward the informal margin. The two labor inputs are

$$L_{F,g} = N_{F,g} \sum_b \theta_b h_b e(h_b), \quad L_{I,g} = \eta_I N_{I,g} h_I e(h_I),$$

where $N_{F,g}$ and $N_{I,g}$ are the headcounts of formal and informal workers in group g (with $N_{I,g} \equiv N_g - N_{F,g}$ and total group employment N_g fixed), h_b indexes the three contracted-hours bins $b \in \{36, 40, 44\}$ with shares θ_b among formal workers, $h_I = 44$ is weekly hours for informal workers, and η_I is the per-head productivity of an informal worker relative to a formal worker. Per-hour efficiency follows $e(h) = \exp\{-\kappa(h - h^*)^2\}$ above the peak $h^* = 40$, where κ is the curvature of the fatigue penalty. Below the peak the paper considers two regimes. The preferred flat-below specification sets $e(h) = 1$ for $h \leq h^*$. The conservative symmetric specification keeps the same quadratic form below h^* . The flat-below specification is informed by Pencavel (2015), who documents the fatigue leg above the peak but does not characterize efficiency below it. The symmetric specification serves as a conservative upper bound on the required TFP gain.

Each firm chooses the number of formal workers $N_{F,g}$ to maximize profit net of three cost wedges,

$$Y_g - \tau_g N_{F,g} - \frac{1}{2} \pi_m N_{I,g}^2 - \frac{1}{2} \gamma_{F,g} (N_{F,g} - \bar{N}_{F,g})^2, \quad (3)$$

where τ_g is a linear per-worker cost of formality (CLT/FGTS obligations and size-dependent enforcement), π_m is a convex cost of informality (fines, reputational risk, and limited access to formal inputs), and $\gamma_{F,g}$ is a quadratic adjustment cost penalizing deviations from the pre-reform formal headcount $\bar{N}_{F,g}$. The wedge τ_g is calibrated to match group-specific informality and plays the role of a size-dependent distortion in the spirit of Hsieh and Klenow (2009) and Restuccia and Rogerson (2008), with endogenous-formality micro-foundations in Dix-Carneiro et al. (2026). Cutting the statutory cap from $h_0 = 44$ hours to h_1 reduces effective labor through both the hours bins h_b and the efficiency function $e(h_b)$, and widens the wedge between the marginal product of formal labor and its private cost.

The object of interest is the TFP multiplier that restores aggregate output,

$$A_{\text{req}} : \sum_g Y_g(A(1 + A_{\text{req}}), K_g, h_1) = \sum_g Y_g(A, K_g, h_0). \quad (4)$$

A_{req} is the one-off percentage gain in aggregate productivity A that exactly offsets the short-run output loss from moving the cap from h_0 to h_1 . It is an accounting benchmark to compare with Brazil’s historical TFP record, not a predicted growth rate.

For welfare, preferences are GHH (Greenwood et al., 1988) with $\sigma_{\text{ghh}} = 1$ and Frisch elasticity $1/\nu = 0.5$. The compensating variation is

$$\Delta\text{CV} = \frac{C_1 - \psi h_1^{1+\nu}/(1+\nu)}{C_0 - \psi h_0^{1+\nu}/(1+\nu)} - 1. \quad (5)$$

The numerator and denominator are the post- and pre-reform consumption-equivalents of welfare under GHH preferences, computed as consumption net of the monetized labor disutility $\psi h^{1+\nu}/(1+\nu)$. ΔCV is the percentage change in welfare.

The model is static and partial-equilibrium. Capital is predetermined, total employment N is fixed, and general-equilibrium feedbacks (wage-price, monetary, fiscal, cross-sector) are absent. The online appendix enumerates the omitted channels and bounds their aggregate effect. A textbook user-cost elasticity $\varepsilon_K \in [0.2, 0.4]$ (Caballero, 1999) with capital share $\alpha = 0.35$ implies

$$\Delta A_{\text{req}}^K \approx -\alpha \cdot \varepsilon_K \cdot |\Delta L_F/L| \in [-1.0, -1.8] \text{ pp}, \quad (6)$$

where ΔA_{req}^K is the downward correction to the short-run A_{req} once the endogenous response of the capital stock is factored in. The medium-run analog of the headline thus falls by 1–2 pp over a 3–5 year horizon.

Table 2 lists the parameters under three headings. External parameters come from PWT, DIEESE, and PNAD Contínua. Moment-matched parameters (τ_g , π_m , κ , ψ) are calibrated internally. Author-chosen parameters (η_I , e_q , h^* , h_I , γ_F^g) are set inside ranges consistent with the work-hours evidence of Pencavel (2015) and Collewet and Sauer-mann (2017), the informal-productivity literature (La Porta and Shleifer, 2014), and the CLT/FGTS compliance-cost literature. Cross-country hours evidence in Bick et al. (2018) provides directional support for declining marginal product of hours but does not identify a peak. The online appendix reports the source and sensitivity for each parameter.

The CES substitution elasticity σ_{sub} is disciplined by the wage premium. From the CES first-order condition, the formal–informal wage premium R is a monotone function of σ_{sub} . Plugging a widened window around Meghir et al.’s estimates, $R \in [1.15, 1.55]$, into the CES FOC at the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$ yields

$$R \in [1.15, 1.55] \implies \sigma_{\text{sub}} \in [1.116, 1.469], \quad A_{\text{req}} \in [7.50\%, 8.58\%] \text{ (conservative, midbox)}. \quad (7)$$

The central value $\sigma = 1.326$ corresponds to $R \approx 1.40$, the midpoint of the MNR win-

Table 2: Parameter values by evidential basis.

Symbol	Description	Value	Source
<i>External</i>			
α	Capital share	0.35	PWT 10.01 + Gollin (2002)
ω	CES formal weight	0.622	1 – narrow informality (PNAD Contínua 2025)
θ_b	Hours-bin weights (formal)	(0.085, 0.269, 0.646)	DIEESE 2024
ν	Inverse Frisch elasticity	2.0	Standard (Frisch = 0.5)
<i>Moment-matched</i>			
σ_{sub}	CES substitution elasticity	1.326 [1.116, 1.469]	Wage premium (Meghir et al., 2015)
τ_S, τ_L	Formal wedges (small, large)	2.609, 0.004	Bisection to group informality
π_m^S, π_m^L	Informality convex cost	0.000, 42.11	Bisection
κ	Efficiency curvature	2.11×10^{-3}	Derived from e_q at h_{ref}
ψ	GHH disutility weight	5.74×10^{-5}	Baseline FOC
<i>Author-chosen (within ranges consistent with evidence)</i>			
η_I	Informal relative efficiency	0.40	La Porta and Shleifer (2014) informal/formal wage ratio midpoint ($\approx 1/3$ to $1/2$)
e_q	Hours–output elasticity	0.60	Pencavel (2015) range
h^*	Peak-efficiency hours	40	Pencavel (2015)
h_I	Informal hours	44	No statutory cap in informal sector
γ_F^S, γ_F^L	Adjustment costs	0.12, 0.03	RAIS + CLT/FGTS

dow.¹ Letting ω and η_I also vary over their data-consistent ranges ($\omega \in [0.58, 0.66]$, $\eta_I \in [0.33, 0.50]$), the joint (σ, ω, η_I) -box envelope under the conservative specification is $[6.93\%, 9.23\%]$ at the eight corners. Under the preferred flat-below specification the central A_{req} is 6.63% and the envelope is $[5.62\%, 7.48\%]$. The online appendix reports the full grids. The small reallocation elasticity documented by Derenoncourt et al. (2025) supports a modest intra-firm σ and is distinct from the high substitution elasticities between worker types reported in refugee studies. The online appendix discusses this distinction.

Rebuilding the calibration on PNAD habitual hours ($\theta = (0.143, 0.406, 0.451)$, 44.2% informality) yields $A_{\text{req}} = 7.80\%$ under the conservative specification. The baseline uses DIEESE contracted hours because the statutory cap is a contracted-hours concept.

3 Validation

Table 3 reports the calibration targets, the domestic alternative-measurement check, and the external directional benchmarks together. Panel A targets are matched by construction. Panel B1 evaluates the model against the PNAD-habitual hours measurement, which the calibration does not target. The implied gap of 1.0 h/week on mean formal hours and 5.8 pp on the share above 36h reflects the contracted-versus-habitual measurement distinction the baseline commits to. Panel B2 reports the Portuguese 1996 reform at the common 44 $\rightarrow$ 40h cap and the Brazilian 1988 reform at its 48 $\rightarrow$ 44h cap. Signs match. Aggregate output falls and hourly productivity rises in both data and model. Magnitudes do not match. The firm-level Portuguese estimate of +4.4% in hourly productivity is about 2.6 points above the aggregate model prediction, a gap consistent with worker selection, work intensification, and sectoral capital deepening that the model does not encode. The Brazilian 1988 evidence points the same way with a slightly larger empirical hourly-wage response. Panel B2 should be read as a directional plausibility benchmark rather than an identified Brazilian lower bound for the effective A_{req} . Portugal and Brazil 1988 differ from the PEC 8/2025 environment in informality, institutional setting, and reform magnitude.

¹The window $R \in [1.15, 1.55]$ spans the full range of MNR’s reported estimates across sub-populations and alternative specifications (Tables 3 and 10). The mapping from R to σ_{sub} depends on (ω, η_I) . The anchors $\omega = 0.622$ (1 minus the narrow PNAD 2025 informality rate) and $\eta_I = 0.40$ (midpoint of the La Porta–Shleifer informal/formal wage ratio range 1/3 to 1/2) are theoretically consistent: under competitive wage-taking the CES aggregator identifies η_I through the marginal-product (wage) ratio rather than through value-added per employee, which is contaminated by capital-per-worker differences. The online appendix documents the derivation.

Table 3: Calibration fit: targets and checks outside calibration.

Moment	Data	Model	Gap
<i>Panel A: Calibration targets (matched by construction)</i>			
Informality rate, small firms (≤ 49 empl.)	0.500	0.500	0.000
Informality rate, large firms	0.200	0.200	0.000
Formal employment share, small firms	0.590	0.590	0.000
Formal employment share, large firms	0.410	0.410	0.000
Aggregate informality, narrow PNAD	0.378	0.377	0.001
Wage premium R , MNR (2015) central target	1.400	1.400	0.000
<i>Panel B1: Domestic checks outside calibration</i>			
Average weekly formal hours (DIEESE vs PNAD habitual)	41.23	42.24	1.01 h/week
Share of formal workers above 36h (DIEESE vs PNAD, %)	85.7	91.5	5.80 pp
Aggregate weekly hours (formal + informal)	39.13	42.91	3.78 h/week
<i>Panel B2: External directional benchmarks</i>			
ΔY at 44 $\rightarrow$ 40h cap, Portugal 1996 (firm-level)	-3.20	-2.00	1.20 pp
$\Delta Y/h$ at 44 $\rightarrow$ 40h cap, Portugal 1996 (%)	+4.40	+1.77	2.63 pp
$\Delta W/h$ at 48 $\rightarrow$ 44h, Brazil 1988 (%), GMC 2003)	+7.00	+1.77	5.23 pp

Panel B1 rows 1–2 exploit the DIEESE-contracted vs PNAD-habitual measurement distinction. Row 3 reports aggregate weekly hours under the $h_I = 44$ assumption, sensitivity-checked in the online appendix (A_{req} shifts by < 0.2 pp for $h_I \in [38, 46]$). Panel B2 compares the model’s aggregate prediction at the 44 $\rightarrow$ 40h cap to the Portuguese firm-level estimate (Asai et al., 2024) and to Brazil’s 1988 reform of comparable cap-cut magnitude (48 $\rightarrow$ 44h, $\sim 8\%$) as documented by Gonzaga et al. (2003). Both external benchmarks suggest the model is conservative on the hourly-productivity response. The model’s $e(h)$ captures fatigue and not worker selection, work intensification, or sectoral capital deepening. Sources: PNAD Contínua Q4 2024, RAIS 2022, Meghir et al. (2015), Asai et al. (2024), Gonzaga et al. (2003).

4 Main results

Table 4 reports the aggregate effects of the reform under both $e(h)$ regimes, holding total employment fixed. Under the preferred flat-below specification, output falls by $\Delta Y = -6.60\%$, aggregate welfare by $\Delta CV = -1.76\%$, and the required TFP gain is $A_{\text{req}} = 6.63\%$. Under the conservative symmetric specification, $\Delta Y = -8.02\%$, $\Delta CV = -3.63\%$, and $A_{\text{req}} = 8.18\%$. Aggregate informality rises by +1.6–1.9 pp under both specifications. With moderate informal relative efficiency ($\eta_I = 0.40$, identified from the observed wage ratio) and mildly substitutable formal–informal labor ($\sigma > 1$), firms shift marginal workers toward informality when the cap tightens instead of absorbing the hours loss intensively. Varying the three load-bearing parameters (σ, ω, η_I) at the corners of their data-consistent ranges gives a joint envelope of $A_{\text{req}} \in [5.62\%, 7.48\%]$ under the preferred specification and $[6.93\%, 9.23\%]$ under the conservative one. The preferred envelope lies strictly below the conservative one because the flat-below fatigue channel contributes positively to per-hour productivity, which the symmetric specification does not deliver. The capital-adjustment bound of Equation (6) lowers the effective A_{req} by 1–1.8 pp at textbook elasticities over a 3–5 year horizon, placing the medium-run analog at roughly 4.8%–5.6% under the preferred specification and 6.4%–7.2% under the conservative one.

Table 4: Main results: 44 $\rightarrow$ 36h reform.		
	Flat-below (preferred)	Symmetric (conservative)
<i>Positive predictions (no TFP compensation)</i>		
Output, ΔY	−6.60%	−8.02%
Welfare (GHH), ΔCV	−1.76%	−3.63%
Informality, ΔInf	+1.57 pp	+1.92 pp
Output per hour, $\Delta Y/h$	+2.39%	+0.75%
<i>Benchmark: TFP gain that restores output</i>		
A_{req} , central	6.63%	8.18%
A_{req} , joint envelope	[5.62, 7.48]%	[6.93, 9.23]%

The joint envelope spans the eight corners of the disciplined (σ, ω, η_I) box. The preferred envelope lies strictly below the conservative one because the flat-below specification has a positive fatigue channel that the symmetric one does not.

The curve in Figure 1 is convex. The first hours of reduction cost little in TFP terms because they move formal workers toward the efficiency peak at $h^* = 40$. Subsequent reductions grow more expensive as the cap overshoots the optimum. The output loss decomposes into a mechanical hours channel (approximately −14.8%, identical across specifications) and a fatigue channel that switches sign between specifications, from small-positive under flat-below to small-negative under symmetric. The CES reallocation residual is positive and moderate. The condition $\sigma > 1$ makes formal and informal labor imperfect substitutes, and the moderate $\eta_I = 0.40$ dampens but does not eliminate the re-

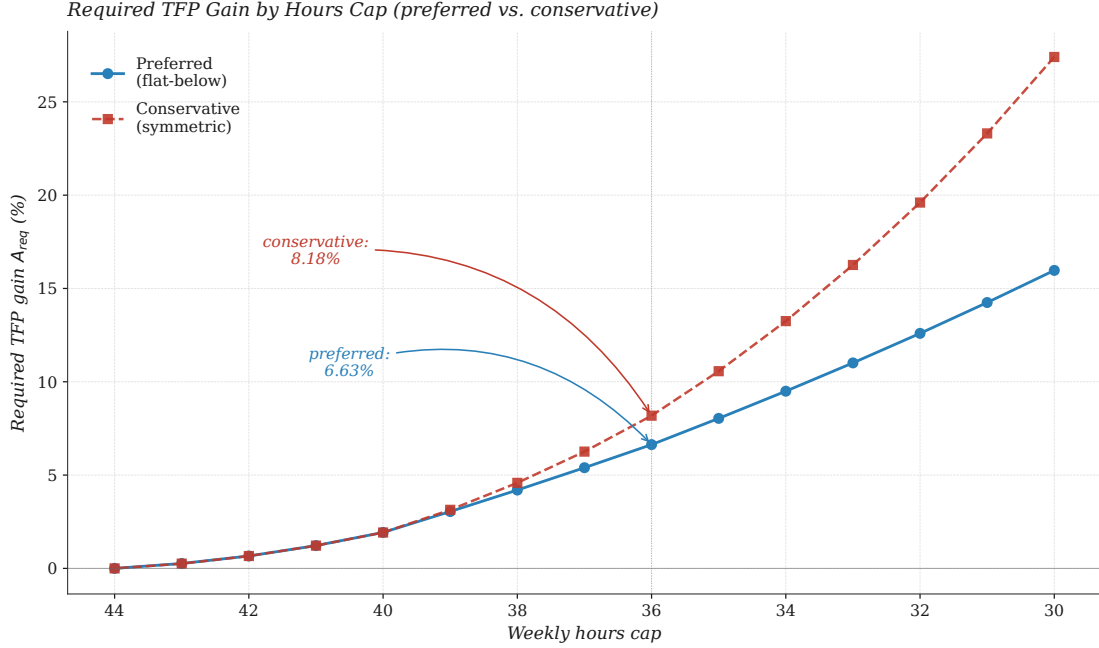


Figure 1: Required TFP gain A_{req} as a function of the weekly hours cap.

Notes: The preferred flat-below schedule yields 6.63% at the proposed 36h cap. The conservative symmetric schedule yields 8.18%. The two schedules coincide above $h^* = 40$ and diverge below it, reflecting different assumptions about $e(h)$ below the efficiency peak.

allocation cost when informality rises. The online appendix reports the full decomposition figure.

We anchor A_{req} against post-1990 Brazilian TFP history using the PWT 10.01 `rtfpna` series (Feenstra et al., 2015). Figure 2 uses the preferred baseline $A_{\text{req}} = 6.63\%$. Brazilian TFP fell by -0.60% per year over 1990–2019 and rose by only $+0.26\%$ per year in the best decade (2003–2013). At the best-decade rate, accumulating the preferred A_{req} takes approximately 25 years and the conservative one takes 31 years. The $+2.3\%$ per year of 1954–1980 reflects the Brazilian economic miracle and is not a realistic reference point. The online appendix reports the full table. The preferred benchmark has no precedent in the post-1990 Brazilian series under any horizon considered.

A pessimistic stress test sets post-reform TFP to $A_{\text{post}} = A(1 - \delta)$ for $\delta \in [0\%, 2\%]$, capturing disruption to learning-by-doing and hysteresis in the informal sector. Table 5 reports the result under the conservative specification. A TFP contraction of 1% raises A_{req} from 8.18% to 9.28% and the welfare loss from 3.63% to 4.89%. A 2% contraction, a magnitude observed in 2015–2016, raises A_{req} to 10.39%. The stress test does not predict a particular δ . It shows that the central prediction does not collapse under moderate post-reform TFP deterioration.

The aggregate compensating variation at the $44 \rightarrow 36$ h reform is -1.76% under the preferred flat-below specification and -3.63% under the conservative symmetric one, both computed from Equation (5). Figure 3 plots the full schedule. Welfare peaks near a 41h

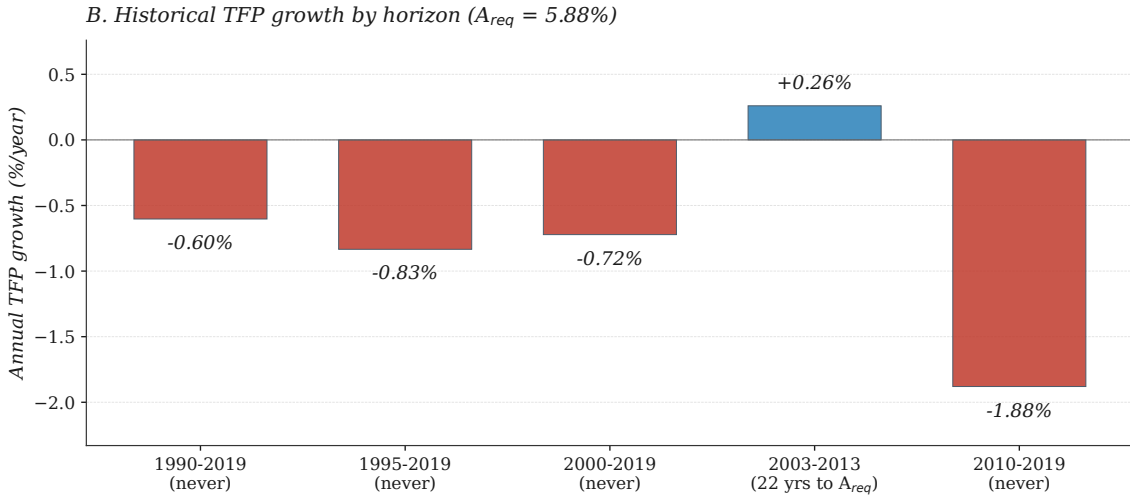
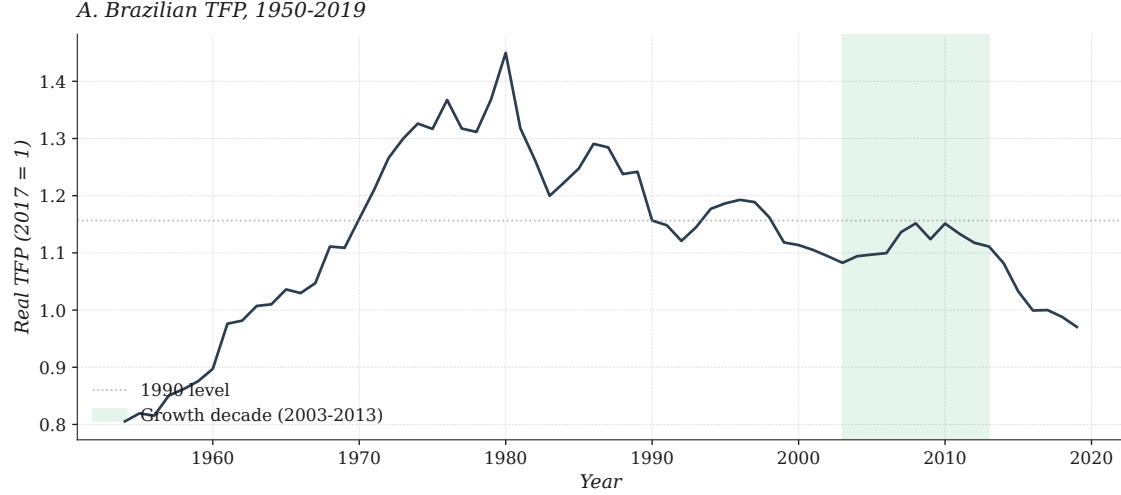


Figure 2: Brazilian TFP and the required one-off productivity gain.
Notes: Panel A plots the PWT `rtfpna` series (real TFP at constant national prices) for Brazil, 1954–2019. Panel B shows annualized TFP growth rates by horizon. Labels indicate the years required to accumulate the preferred baseline $A_{req} = 6.63\%$ at each historical rate (“never” when the rate is non-positive). *Source:* Penn World Table 10.01 (Feenstra et al., 2015).

Table 5: Pessimistic stress test: reform with post-reform TFP shock.

Post-reform TFP shock δ (%)	A_{req} (%)	ΔY (%)	ΔInf (pp)	ΔCV (%)
0.0 (baseline)	+8.18	−8.02	+1.917	−3.63
0.5	+8.73	−8.51	+1.980	−4.26
1.0	+9.28	−9.00	+2.044	−4.89
1.5	+9.83	−9.50	+2.122	−5.54
2.0	+10.39	−9.99	+2.185	−6.17

δ represents a post-reform contraction of TFP beyond the mechanical effect of the hours cap. The $\delta = 0$ row reproduces the main result of Table 4. Other rows show how outcomes deteriorate if the reform itself depresses TFP by δ .

cap at approximately +0.80% under both specifications, because the leisure gain outweighs the consumption loss in that range. The two schedules coincide between 44h and 40h and diverge below 40h. The coincidence reflects a structural feature. At any cap in $[40, 44]$ the binding workers come from the 44h bin, and their per-hour efficiency rises toward the peak under both specifications (from $e(44) = 0.967$ toward $e(40) = 1$). The divergence below 40h reflects the difference in specification. The flat-below case keeps $e(h) = 1$ for $h \leq 40$, while the symmetric case assigns an efficiency penalty that accumulates as the cap moves further below the peak. Welfare turns negative below 38h in both specifications. A moderate reduction from 44 to 41 raises welfare inside the model. The full $44 \rightarrow 36$ reform does not.

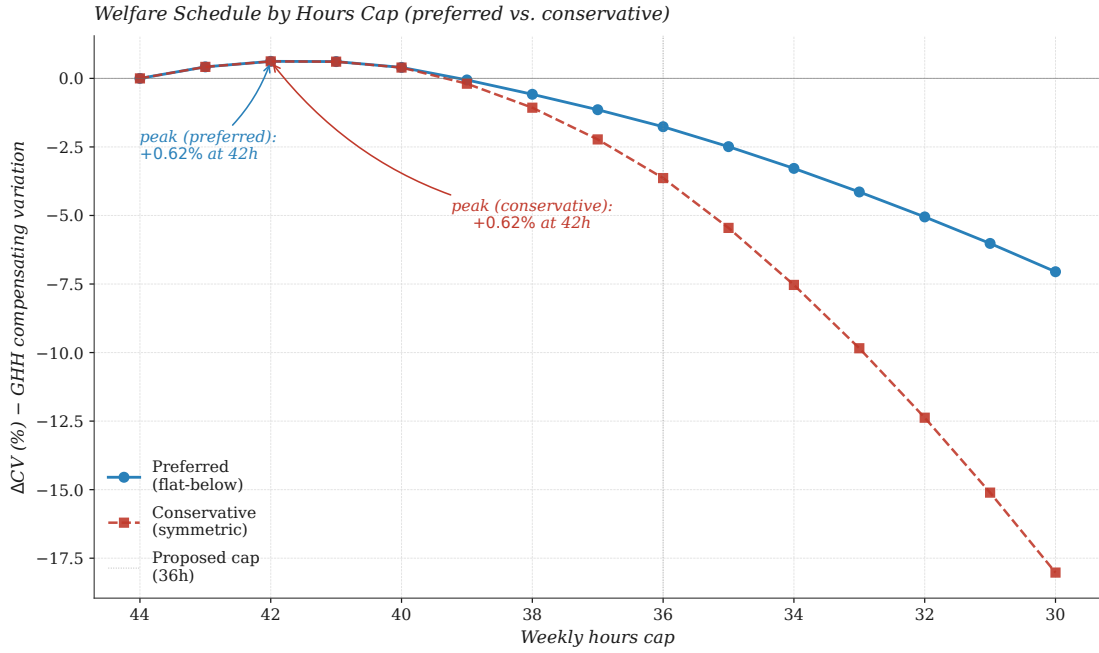


Figure 3: Welfare schedule by hours cap under both calibrations.

Notes: Each line plots aggregate welfare ΔCV from Equation (5) at successively lower hours caps, relative to the 44h baseline, under the preferred flat-below and conservative symmetric $e(h)$ specifications. ΔCV peaks near 41h at about +0.80% and turns negative below 38h. At the proposed 36h cap, welfare falls by 1.76% under the preferred specification and by 3.63% under the conservative one.

A TFP gain of approximately 1.4% at the 36h cap is enough to make the reform welfare-neutral under the preferred specification. The conservative threshold is 2.9%. Both thresholds lie below the output-neutral A_{req} because leisure partially compensates the consumption loss. Welfare-neutral and output-neutral TFP differ by construction. The online appendix reports a type-level decomposition and reconciles the GHH-nonlinearity gap between type-level and representative-agent CV.

Adjustment costs and informality differ across firm sizes and sectors, producing heterogeneous incidence of the reform. Small firms (≤ 49 employees) require a larger TFP gain than large firms, reflecting their higher formalization wedges and baseline informality.

Applying the national framework separately to three broad sectors (agriculture, industry, services) with sector-specific PNAD parameters delivers a per-sector A_{req} of 7.60% in agriculture, 8.05% in industry, and 6.91% in services under the conservative specification, with aggregate informality rising in every sector (agriculture +2.88 pp, industry +2.73 pp, services +2.10 pp). Industry’s A_{req} slightly exceeds agriculture’s because the moderate $\eta_I = 0.40$ combined with industry’s larger formal base dampens the mechanical gap between the two sectors. Services account for most of the aggregate output loss through their output share, not through a larger per-worker effect. The online appendix reports the full decomposition and the parameter sensitivities.

5 Policy implications

The paper quantifies a trade-off the Brazilian policy debate has so far left implicit. Cutting the statutory workweek from 44 to 36 hours affects the majority of Brazilian formal workers and reduces the labor input of the economy by about 14% mechanically. Whether the economy absorbs this cut without an output loss depends on the size of the simultaneous productivity gain. The paper puts a number on that gain. The aggregate output loss is offset by a one-off productivity gain of 6.63% (preferred specification) to 8.18% (conservative upper bound). Neither figure has precedent in the post-1990 Brazilian productivity record. In the best decade after the Real Plan, productivity rose by about a quarter of a percent per year. Matching what the reform requires at that rate would take twenty-five to thirty-one years.

Four messages follow from the model.

First, the 36-hour proposal lies outside the post-1990 Brazilian productivity record. The reform would require a one-off productivity gain that the country has not produced over three decades. Without that gain, the model predicts a short-run output loss of 6.6 to 8.0% and an aggregate welfare loss of 1.8 to 3.6%. The productivity requirement is a benchmark, not a forecast of what TFP will do.

Second, a smaller cap is welfare-improving inside the model. Cutting the workweek from 44 to 41 hours raises aggregate welfare because the value workers place on extra leisure exceeds the consumption they give up. Below 41 hours welfare falls. Below 38 it is negative. The welfare-optimal cap inside the model is near 41 hours. A cap of 36 hours overshoots that interior optimum. The interior optimum is internal to the model and does not constitute a policy recommendation.

Third, the cost of the reform is concentrated. Small firms, which employ most formal workers in Brazil, bear the largest per-worker productivity requirement. Industry and agriculture carry the largest per-sector requirements, and informality rises in all three sectors rather than falling. Services, although not the most exposed per worker, account for the bulk of the aggregate output loss through scale. Policy design that ignores this

concentration will understate the adjustment burden.

Fourth, the reallocation to informality is material but second-order to the hours channel. The calibrated model, disciplined by the observed formal–informal wage premium, finds an aggregate informality increase of 1.6–1.9 pp. This informalization is material for welfare accounting and small relative to the roughly 15% mechanical hours contraction. The output cost comes primarily from fewer hours worked. The reform also pushes additional workers into informality, weakening its own formalization goals.

These conclusions come with limits. The model is short-run and partial-equilibrium. Endogenous capital adjustment over a three-to-five-year horizon lowers the required productivity gain by about one to two percentage points. Total employment is held fixed, which shuts down work-sharing. The empirical evidence on work-sharing from Germany (Hunt, 1996) and Brazil’s 1988 constitutional reform (Gonzaga et al., 2003) suggests this restriction is defensible but not innocuous. The Portuguese 1996 experience (Asai et al., 2024) documents firm-level productivity gains larger than the model produces, which suggests the model is conservative on the hourly-productivity response.

Subject to these limits, the model-implied implication is direct. The 44-to-36 reform as currently proposed would require a one-off TFP gain outside the post-1990 Brazilian record. A phased path with an intermediate stop near 41 hours is more consistent with the model’s welfare schedule and would buy time for the capital stock, the informal margin, and firm-level productivity to adjust.

Online appendix. An online appendix containing data sources, partial-equilibrium omissions, calibration details, sensitivity grids, sectoral decomposition, type-level welfare, output decomposition, the historical TFP table, derivations, and the asymmetric- $e(h)$ robustness is available at [URLpending].

Code and data availability. Code and calibration targets to reproduce every numbered table and figure in this paper are available at the author’s public repository <https://github.com/vr-rodriques/jornada-6x1-replication> (to be activated upon publication).

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